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MODEL SELECTION AND VALIDATION REPORT ( fishing_tric model )
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## OBSERVATIONS:1755

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( fishing_tric model )

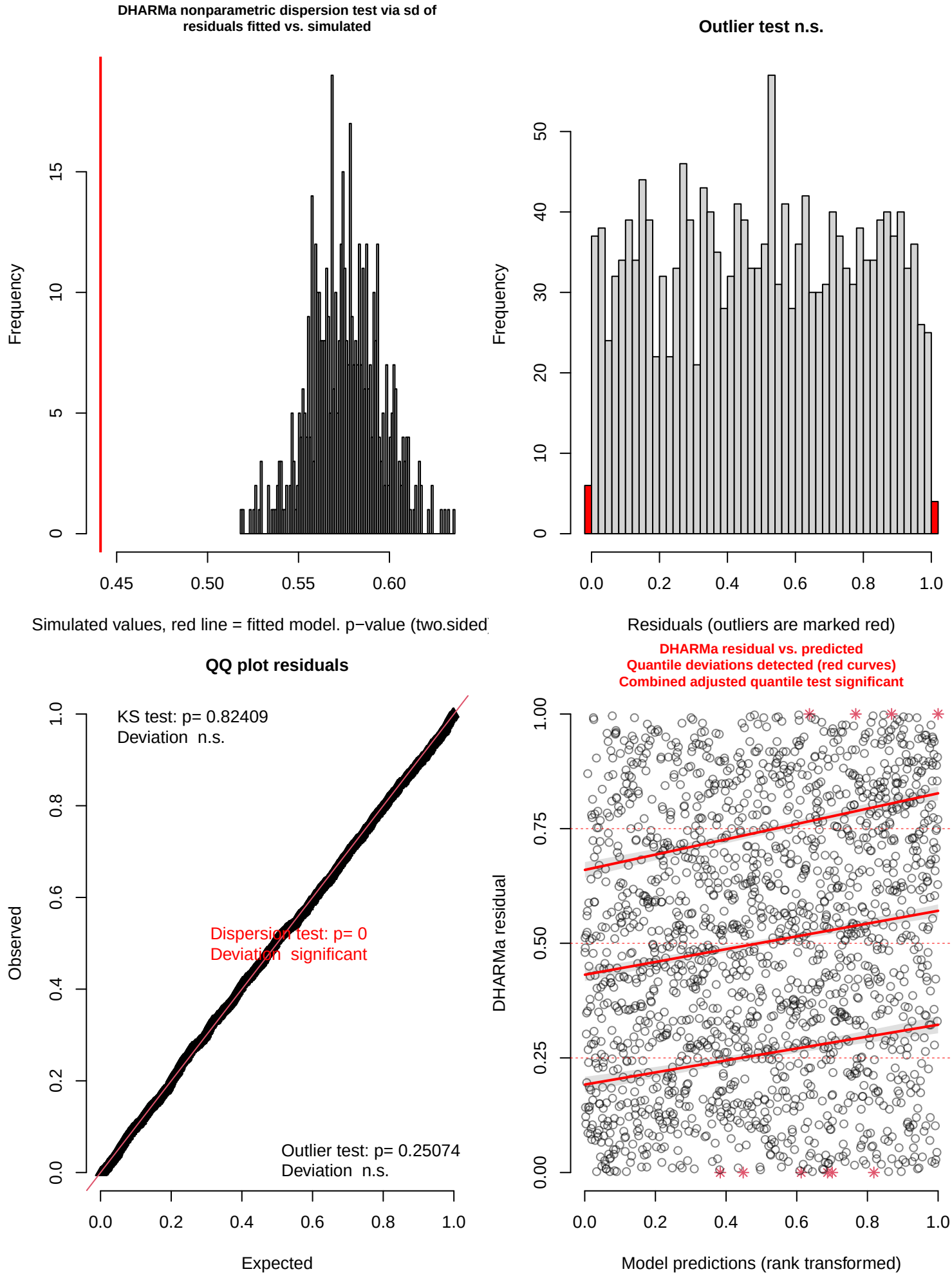
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: gaussian(link = 'identity')

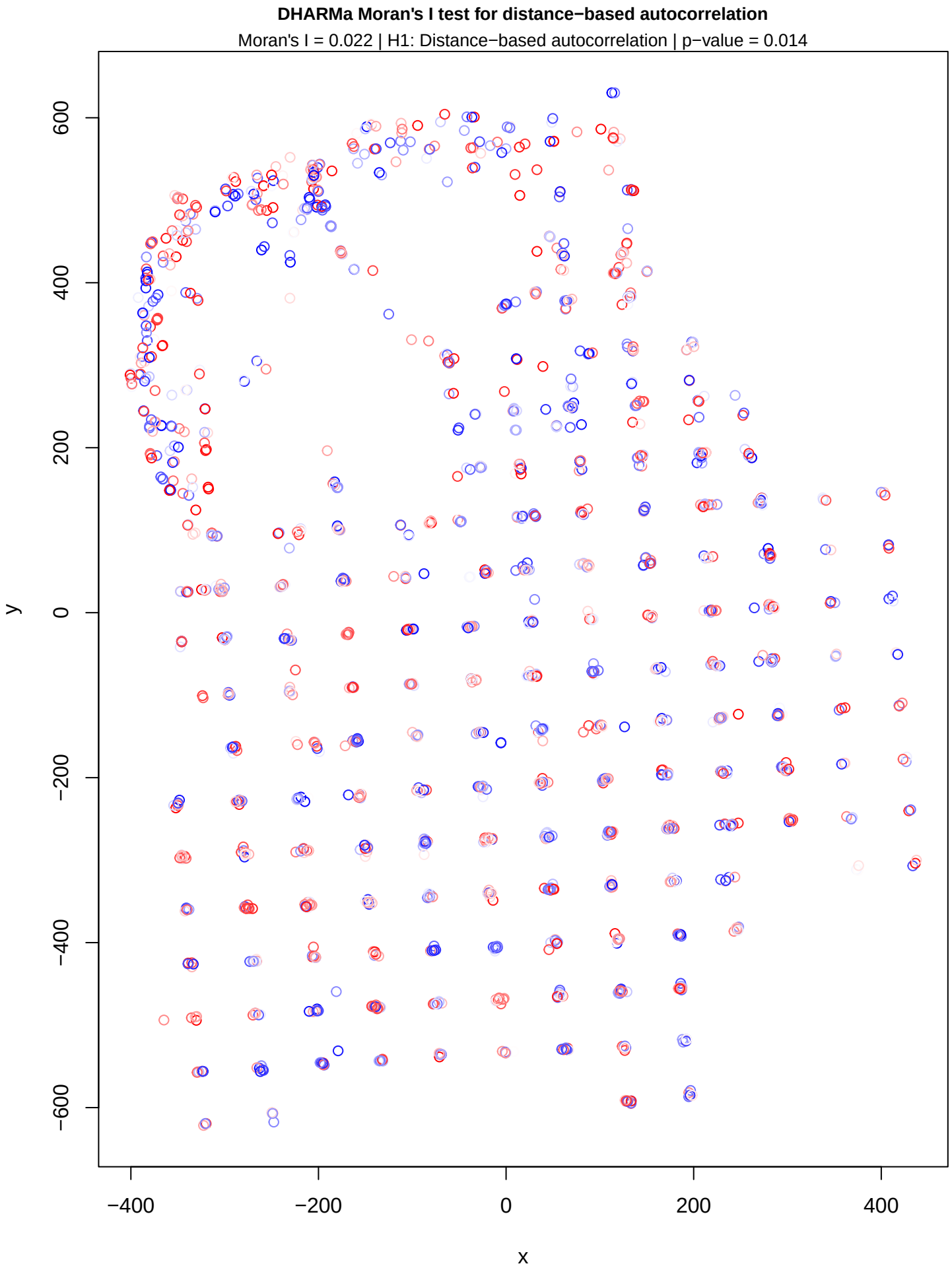
Conditional model:
                coef.est coef.se
(Intercept)        −0.08    0.48
log_sum_fishing     0.00    0.02
sbt_mean            0.33    0.09
sbt_sd              0.02    0.04
sst_sd             −0.08    0.04
sic_mean            0.03    0.06
log_chla_mean       0.01    0.03
depth              −0.05    0.04

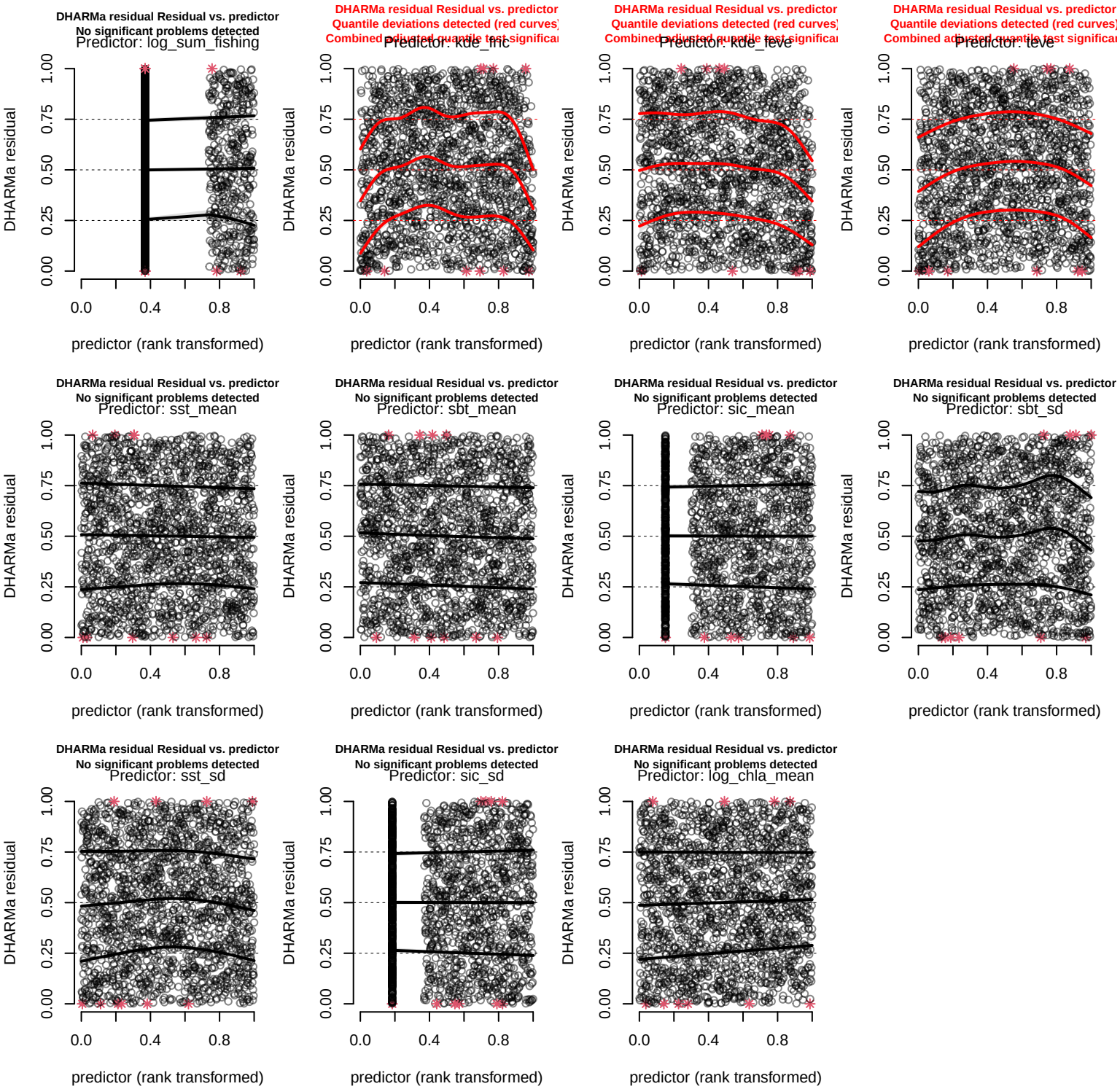
Dispersion parameter: 0.72
Spatiotemporal AR1 correlation (rho): 0.96
Matérn range: 448.40
Spatiotemporal marginal AR1 SD: 1.09
REML criterion at convergence: 2154.009

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                  df      AIC   delta_AIC
spatiotemporal_random_fields_ar1                  12 4332.018   0.7787530
spatiotemporal_random_fields_ar1_with_anisotropy 14 4332.797 193.5537680
spatial_random_fields_with_anisotropy             13 4526.350   0.6909055
spatial_random_fields                             11 4527.041  36.7388577
spatiotemporal_random_fields_iid                  11 4563.780   3.4145324
spatiotemporal_random_fields_iid_with_anisotropy 13 4567.195 303.8518611
error_only                                         9 4871.047          NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
0.45
0.50
0.55
0.60
Simulated values, red line = fitted model. p−value (two.sided
Outlier test n.s.
Frequency
0
10
20
30
40
50
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.82409
Deviation n.s.
Dispersion test: p= 0
Deviation significant
Outlier test: p= 0.25074
Deviation n.s.
Observed
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
0.00
0.25
0.50
0.75
1.00
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.022 | H1: Distance−based autocorrelation | p−value = 0.014
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. predictor
No significant problems detected
Predictor: log_sum_fishing
DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significan
Predictor: kde_fric
Predictor: kde_feve
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8